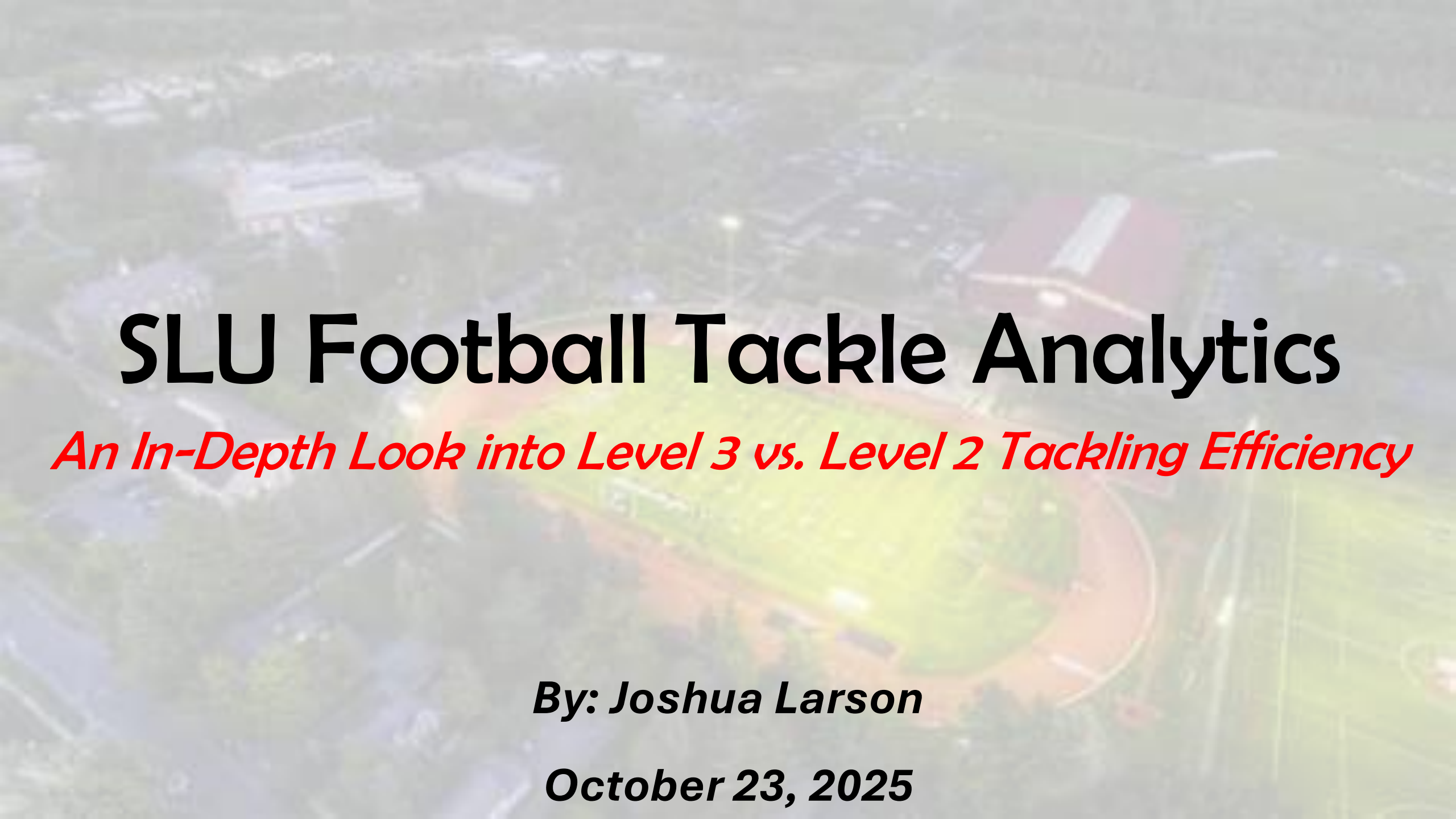


An aerial, slightly blurred photograph of a large football stadium. The green pitch is visible in the center, surrounded by tiered seating areas that appear to be reddish-brown. The stadium is situated in an urban or suburban area with some surrounding buildings and greenery visible in the background.

# SLU Football Tackle Analytics

*An In-Depth Look into Level 3 vs. Level 2 Tackling Efficiency*

***By: Joshua Larson***

***October 23, 2025***

# What We Will Cover...

## Quick recap

- Grading Scheme
- Data Collection Process
- Data Assembly

## Updated

- Manipulations
- Visualizations
- Conclusion

# Grading Scheme

## 0-4 Values

- 0: Non-existent, whiff
- 1: Effort made, slight action made
- 2: Bare minimum of okay. Did the profile to an acceptable degree
- 3: Great play in the aspect, effective and optimal for a good tackle
- 4: Perfect, teach tape

**Track**: Evaluates angle, leverage, and speed in closing space

**Prepare**: Focus on base, pad level, head positioning, and footwork before contact

**Connect**: Eyes up and winning with point-on-point contact

**Accelerate**: Maintaining momentum by running feet and hips through

**Finish**: The control in a wrap, roll, or through motion all the way to the ground

# The Other Variables

**Base Columns:** PLAY #, ODK, DN, DIST, YARD LN, HASH, GN/LS, PERSONNEL, PLAY TYPE, PLAY RESULT, DEF. PLAY CALL

**Tackle Columns:** TACKLER (Jersey #), TACKLE POS (DL, DE, LB, etc.), TACKLE ENTRY (Front, Side, Rear), LEVEL (3, 2, 1), ENGAGED, MAKE/MISS, YAFC (first chance/contact, ASSISTED, FIELD\_TYPE (Box, Alley, Open, Sideline)

# The Data

- How is the data collected?
  - Hudl Sideline → Hudl
  - Cutup and Tag
  - In-depth watch on every play with a tackle
  - Only grade first chance tackle attempts
  - Collected leveraging data simultaneously
  - Export to .xlsx

# Loading the Dataset

```
import pandas as pd

# 1) Load
utica = pd.read_excel("../Python_data/UticaTackleData_2025-08-23.xlsx")
moville = pd.read_excel("../Python_data/MorrisvilleTackleData_2025-08-30.xlsx")
norwich = pd.read_excel("../Python_data/NorwichTackleDataV2_2025-09-11.xlsx")
sjf = pd.read_excel("../Python_data/SJFTackleDataV2_2025-09-17.xlsx")
alfred = pd.read_excel("../Python_data/AlfredTackleDataV2_2025-09-20.xlsx")
union = pd.read_excel("../Python_data/UnionTackleDataV2_2025-09-27.xlsx")
hilbert = pd.read_excel("../Python_data/HilbertTackleDataV2_2025-10-04.xlsx")
ithaca = pd.read_excel("../Python_data/IthacaTackleDataV2_2025-10-11.xlsx")
hobart = pd.read_excel("../Python_data/HobartTackleDataV2_2025-10-25.xlsx")
rpi = pd.read_excel("../Python_data/RPITackleDataV2_2025-11-01.xlsx")
buffstate = pd.read_excel("../Python_data/BufStateTackleDataV2_2025-11-08.xlsx")
rochester = pd.read_excel("../Python_data/RochesterTackleDataV2_2025-11-15.xlsx")

# 2) Tag game and date
utica["GAME"], utica["GAME_DATE"], utica['WEEK'] = "UT", "2025-08-23", "-1"
moville["GAME"], moville["GAME_DATE"], moville['WEEK'] = "MO", "2025-08-30", '0'
norwich["GAME"], norwich["GAME_DATE"], norwich['WEEK'] = "NR", "2025-09-05", '1'
sjf["GAME"], sjf["GAME_DATE"], sjf['WEEK'] = "SJF", "2025-09-13", '2'
alfred["GAME"], alfred["GAME_DATE"], alfred['WEEK'] = "ALF", "2025-09-20", '3'
union["GAME"], union["GAME_DATE"], union['WEEK'] = "UC", "2025-09-27", '4'
hilbert["GAME"], hilbert["GAME_DATE"], hilbert['WEEK'] = "HC", "2025-10-04", '5'
ithaca["GAME"], ithaca["GAME_DATE"], ithaca['WEEK'] = "IC", "2025-10-11", '6'
hobart["GAME"], hobart["GAME_DATE"], hobart['WEEK'] = "HOB", "2025-10-25", '7'
rpi["GAME"], rpi["GAME_DATE"], rpi['WEEK'] = "RPI", "2025-11-01", '8'
buffstate["GAME"], buffstate["GAME_DATE"], buffstate['WEEK'] = "BSU", "2025-11-08", '9'
rochester["GAME"], rochester["GAME_DATE"], rochester['WEEK'] = "UOR", "2025-11-15", '10'

# 3) Combine
tackle = pd.concat([utica, moville, norwich, sjf, alfred, union, hilbert, ithaca, hobart, rpi, buffstate, rochester], ignore_index=True)

tech_cols = ["TRACK", "PREPARE", "CONNECT", "ACCELERATE", "FINISH"]
tackle["TECH_SCORE"] = tackle[tech_cols].sum(axis=1)
tackle_clean = tackle.loc[(tackle["PLAY_RESULT"] != "PENALTY") & (tackle["LEVEL"] != 0)]
tackle = tackle_clean
tackle
```

- Packages used:

- Pandas

- Matplotlib

- Numpy

- Plotnine

- Scipy

- Statsmodels

# Standardizing the Dataset

- Added the TECH\_SCORE variable by summing the technique columns.
- Initiated tackle = tackle\_clean for all manipulations: 557 → 542
  - Several large plays with YAFC>20 were on PENALTIES
  - Removed tackle Level 0

```
tech_cols = ["TRACK", "PREPARE", "CONNECT", "ACCELERATE", "FINISH"]
tackle["TECH_SCORE"] = tackle[tech_cols].sum(axis=1)
tackle_clean = tackle.loc[(tackle["PLAY RESULT"] != "PENALTY") & (tackle["LEVEL"] != 0)]
tackle = tackle_clean
tackle
```

# Base Analyses: Tackle Level & MAKE/MISS

## Tackle Count by Level

- Total = 542

| LEVEL | count |
|-------|-------|
| 3     | 342   |
| 2     | 145   |
| 1     | 55    |

## YAFC By Level

- Total = 1853

- Average YAFC = 3.84

| LEVEL | YAFC |
|-------|------|
| 1     | 260  |
| 2     | 366  |
| 3     | 1455 |

## MAKE/MISS

- 334 MAKE, 208 MISS

- AVG YAFC on MAKE: 1.57, MISS: 7.49

| MAKE/MISS | proportion |
|-----------|------------|
| MAKE      | 0.616236   |
| MISS      | 0.383764   |

- MAKE/MISS by Tackle Level

| LEVEL | MAKE/MISS | proportion |
|-------|-----------|------------|
| 1     | MAKE      | 0.618182   |
|       | MISS      | 0.381818   |
| 2     | MAKE      | 0.717241   |
|       | MISS      | 0.282759   |
| 3     | MAKE      | 0.573099   |
|       | MISS      | 0.426901   |

## Average YAFC Allowed on MAKE/MISS by Level

| LEVEL | MAKE/MISS | YAFC     |
|-------|-----------|----------|
| 1     | MAKE      | 2.000000 |
|       | MISS      | 9.142857 |
| 2     | MAKE      | 1.221154 |
|       | MISS      | 5.829268 |
| 3     | MAKE      | 1.673469 |
|       | MISS      | 7.719178 |

# Base Analyses: Tackle Entry

## Tackle Count by Entry

| TACKLE ENTRY | Count |
|--------------|-------|
| SIDE         | 283   |
| FRONT        | 181   |
| REAR         | 78    |

## Average YAFC by Entry

| TACKLE ENTRY | YAFC     |
|--------------|----------|
| FRONT        | 2.928177 |
| REAR         | 3.782051 |
| SIDE         | 4.438163 |

## MAKE/MISS % by Entry

| TACKLE ENTRY | MAKE/MISS | proportion |
|--------------|-----------|------------|
| FRONT        | MAKE      | 0.629834   |
|              | MISS      | 0.370166   |
| REAR         | MAKE      | 0.756410   |
|              | MISS      | 0.243590   |
| SIDE         | MAKE      | 0.568905   |
|              | MISS      | 0.431095   |

## Average Technique Score by Entry

| TACKLE ENTRY | TECH_SCORE |
|--------------|------------|
| FRONT        | 9.176796   |
| REAR         | 9.282051   |
| SIDE         | 7.904594   |

# Top Tacklers: Minimum 10 Attempts

Most First-Chance Attempts

| TACKLER | ↕ | <u>123</u> TACKLER | ↕ |
|---------|---|--------------------|---|
| 15      |   | 67                 |   |
| 52      |   | 44                 |   |
| 31      |   | 42                 |   |
| 20      |   | 35                 |   |
| 3       |   | 33                 |   |
| 21      |   | 32                 |   |
| 48      |   | 29                 |   |
| 35      |   | 28                 |   |
| 28      |   | 26                 |   |
| 47      |   | 24                 |   |
| 75      |   | 20                 |   |
| 27      |   | 17                 |   |
| 94      |   | 14                 |   |
| 56      |   | 12                 |   |
| 55      |   | 11                 |   |

Highest Average TECH\_SCORE

| TACKLER | ↕ | <u>123</u> TECH_SCORE | ↕ |
|---------|---|-----------------------|---|
| 52      |   | 10.272727             |   |
| 3       |   | 10.181818             |   |
| 31      |   | 10.000000             |   |
| 47      |   | 9.666667              |   |
| 48      |   | 9.448276              |   |
| 94      |   | 9.357143              |   |
| 21      |   | 9.343750              |   |
| 28      |   | 9.192308              |   |
| 20      |   | 8.885714              |   |
| 15      |   | 8.835821              |   |
| 27      |   | 8.529412              |   |
| 55      |   | 6.363636              |   |
| 75      |   | 6.350000              |   |
| 35      |   | 5.428571              |   |
| 56      |   | 3.750000              |   |

Top First Chance Tackle Make %

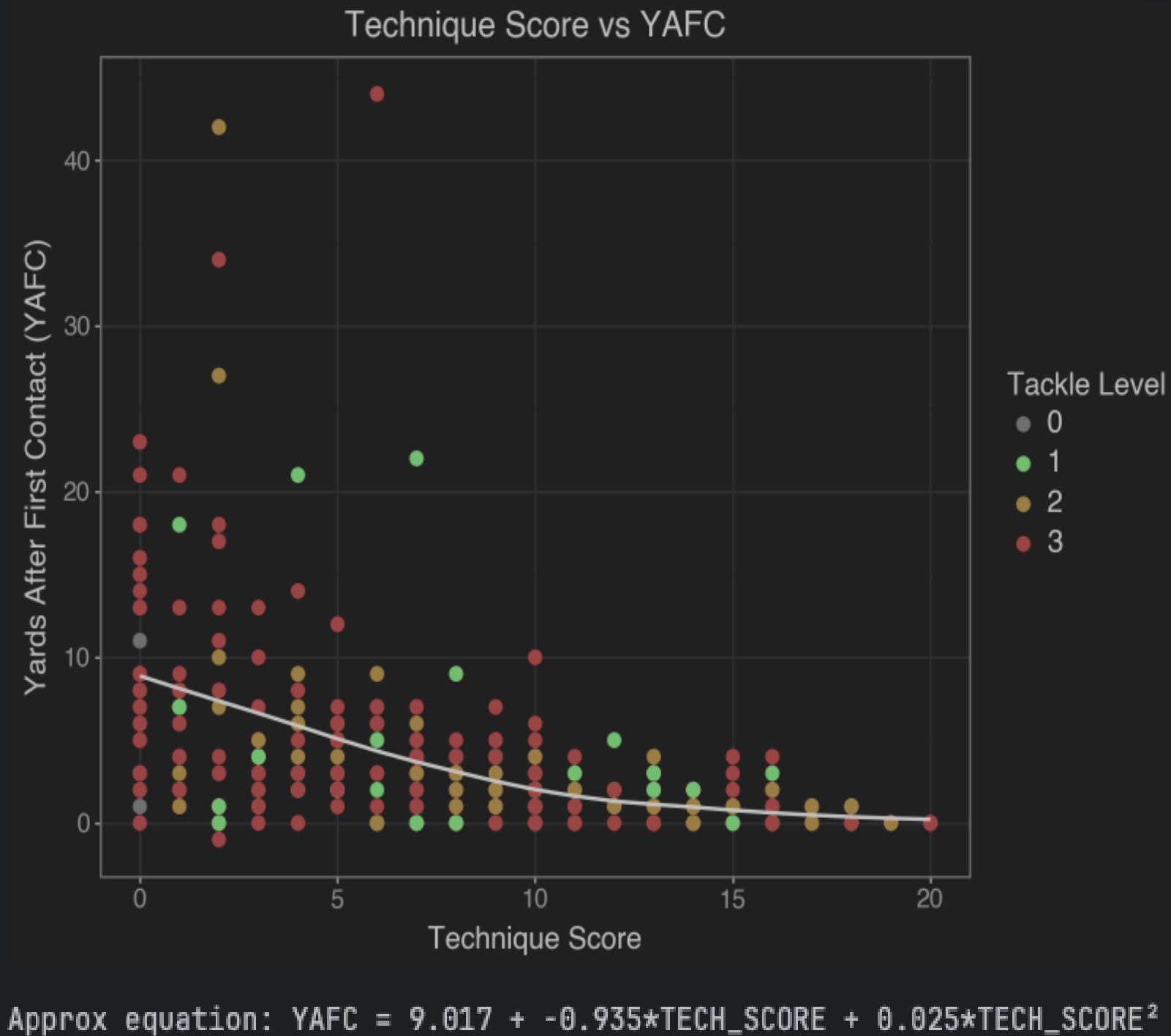
| TACKLER | ↕ | <u>123</u> MAKE/MISS | ↕ |
|---------|---|----------------------|---|
| 94      |   | 0.857143             |   |
| 97      |   | 0.777778             |   |
| 21      |   | 0.750000             |   |
| 47      |   | 0.750000             |   |
| 28      |   | 0.692308             |   |
| 52      |   | 0.681818             |   |
| 3       |   | 0.666667             |   |
| 24      |   | 0.666667             |   |
| 48      |   | 0.655172             |   |
| 31      |   | 0.642857             |   |
| 20      |   | 0.571429             |   |
| 15      |   | 0.567164             |   |
| 27      |   | 0.529412             |   |
| 75      |   | 0.500000             |   |
| 35      |   | 0.500000             |   |

Lowest AVG YAFC

| TACKLER | ↕ | <u>123</u> YAFC | ↕ |
|---------|---|-----------------|---|
| 94      |   | 2.285714        |   |
| 48      |   | 2.620690        |   |
| 21      |   | 2.812500        |   |
| 52      |   | 2.954545        |   |
| 55      |   | 3.363636        |   |
| 3       |   | 3.393939        |   |
| 20      |   | 3.457143        |   |
| 47      |   | 3.791667        |   |
| 15      |   | 3.880597        |   |
| 31      |   | 3.880952        |   |
| 27      |   | 3.941176        |   |
| 28      |   | 4.346154        |   |
| 75      |   | 5.250000        |   |
| 35      |   | 5.571429        |   |
| 56      |   | 5.916667        |   |

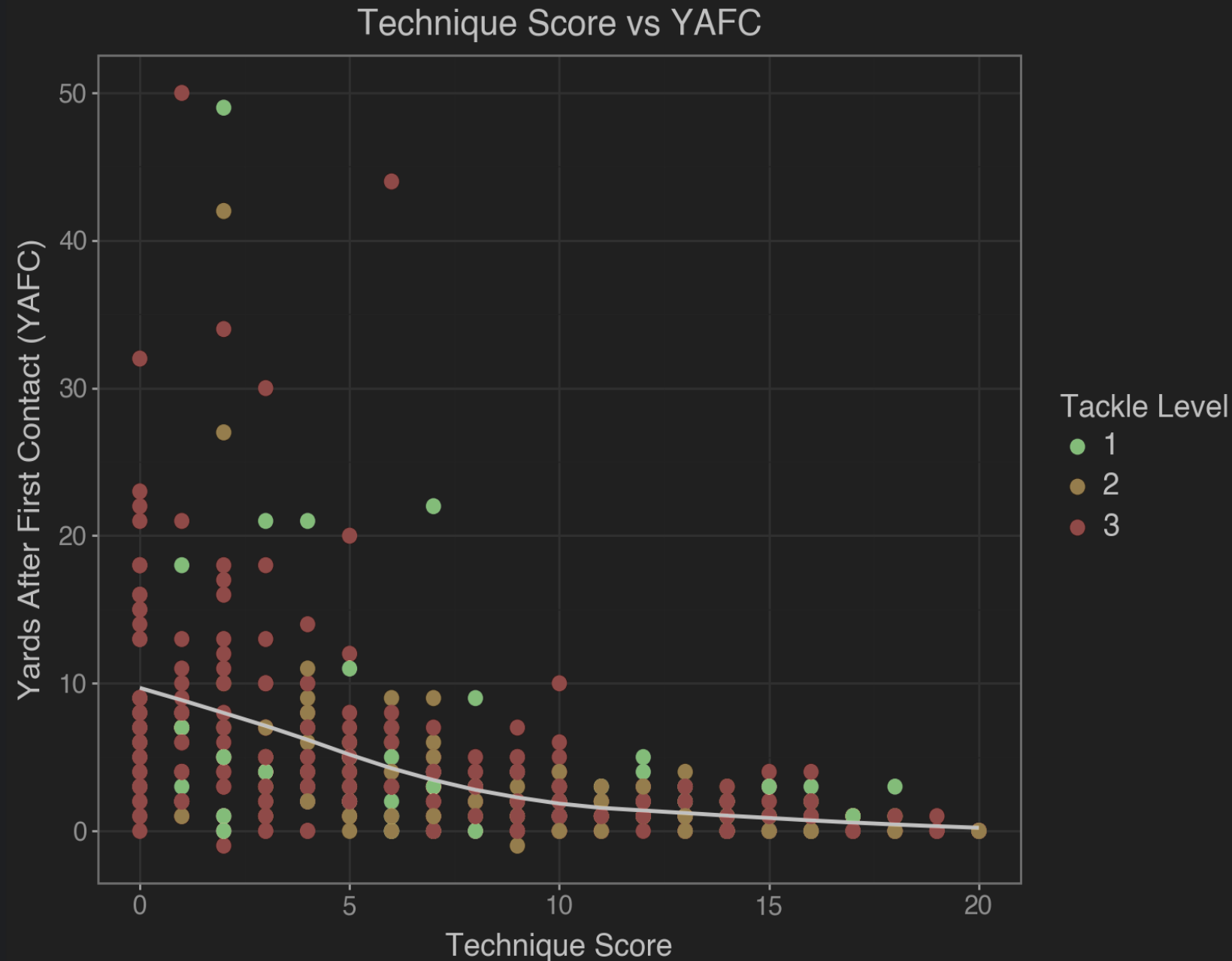
# Visualizations

- Fit a 2<sup>nd</sup>-order Polynomial
- Has a quadratic shape; any others would overfit or be too noisy
- Manually coded because Plotnine doesn't supply fitted equations.



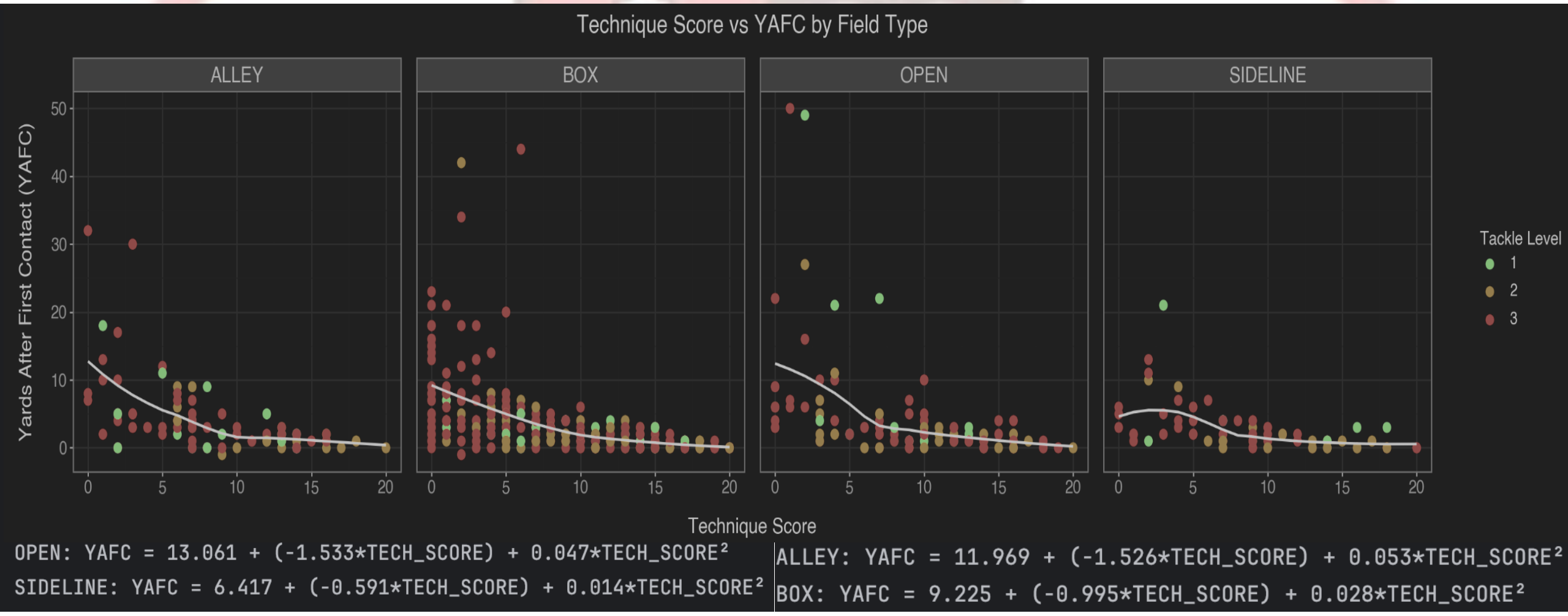
# Visualizations

- Scatterplot of Technique Score vs. Yards After First Contact



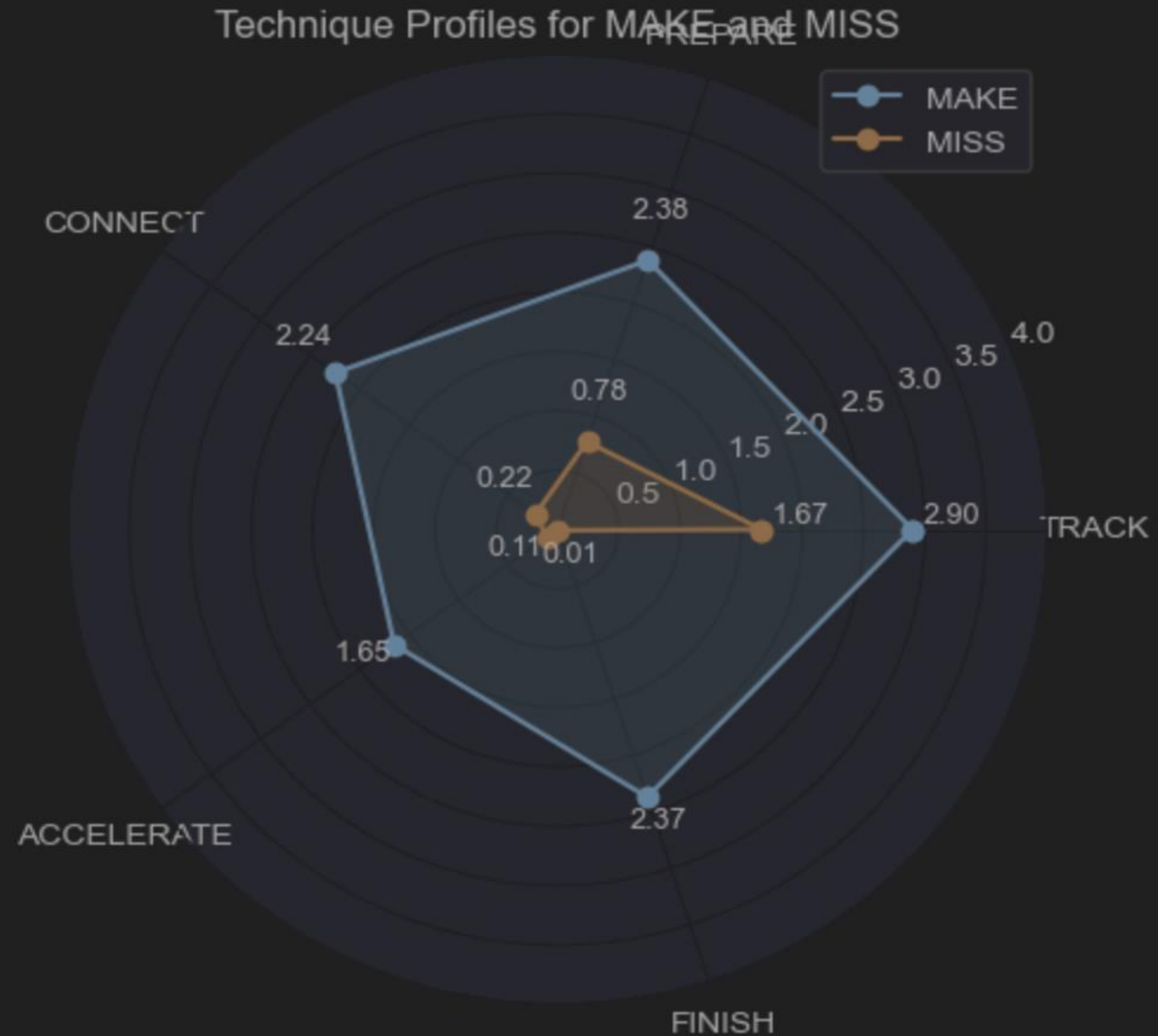
Approx equation:  $YAFC = 9.893 + -1.096*TECH\_SCORE + 0.032*TECH\_SCORE^2$

# Visualizations: Technique Score vs YAFC by Field Type



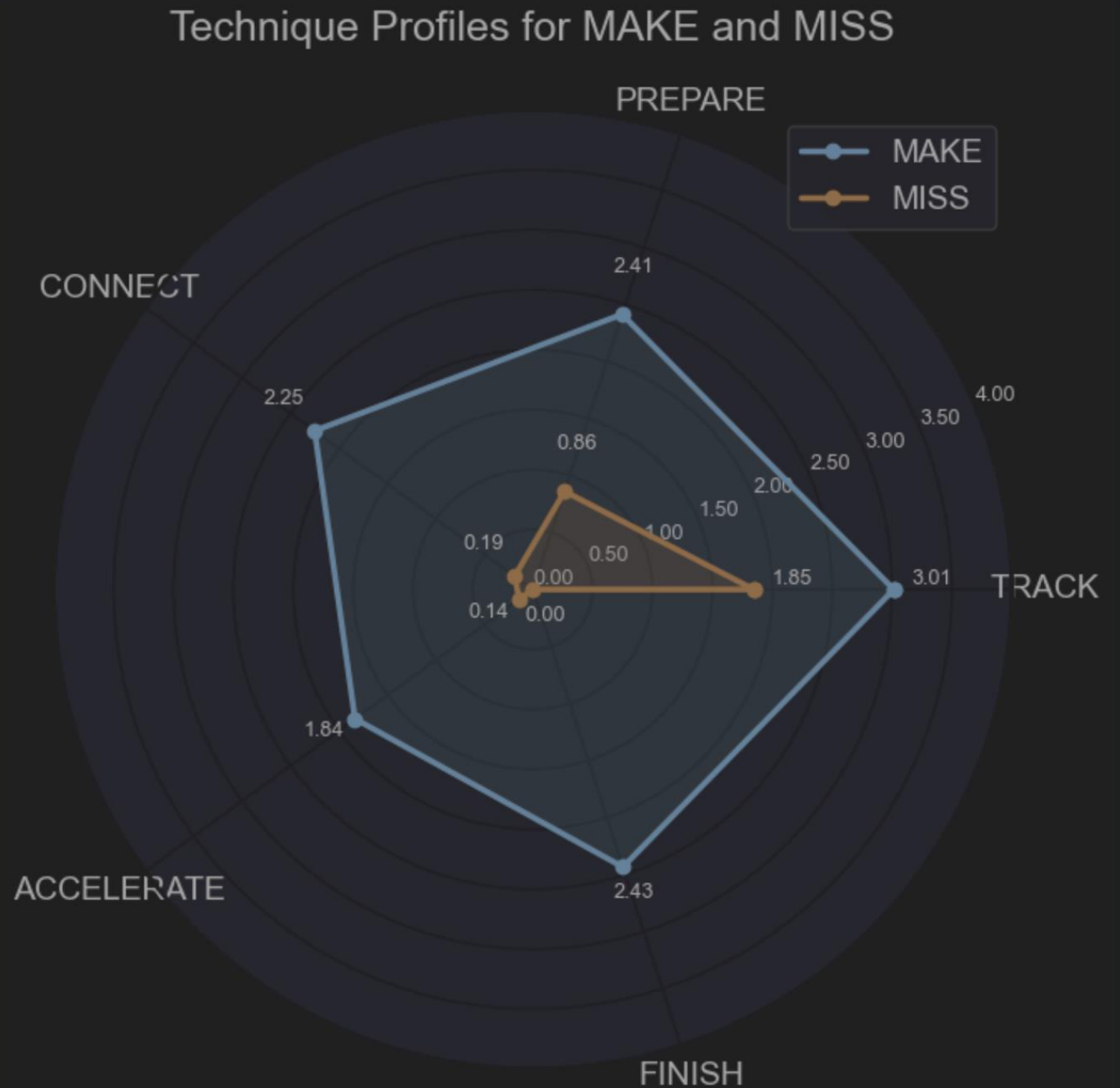
# Visualizations

- Radar Chart showing Averages of Technique Profiles from Week -1:6



# Visualizations

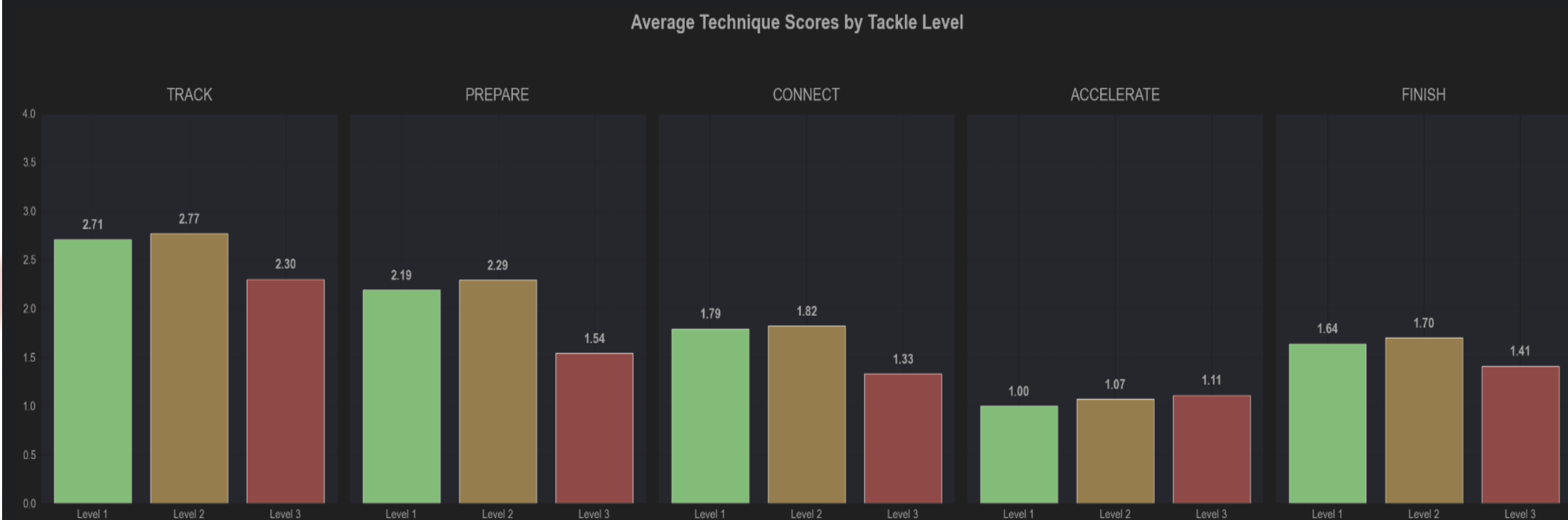
- Radar Chart showing  
Averages of Technique  
Profiles from Week -1:10



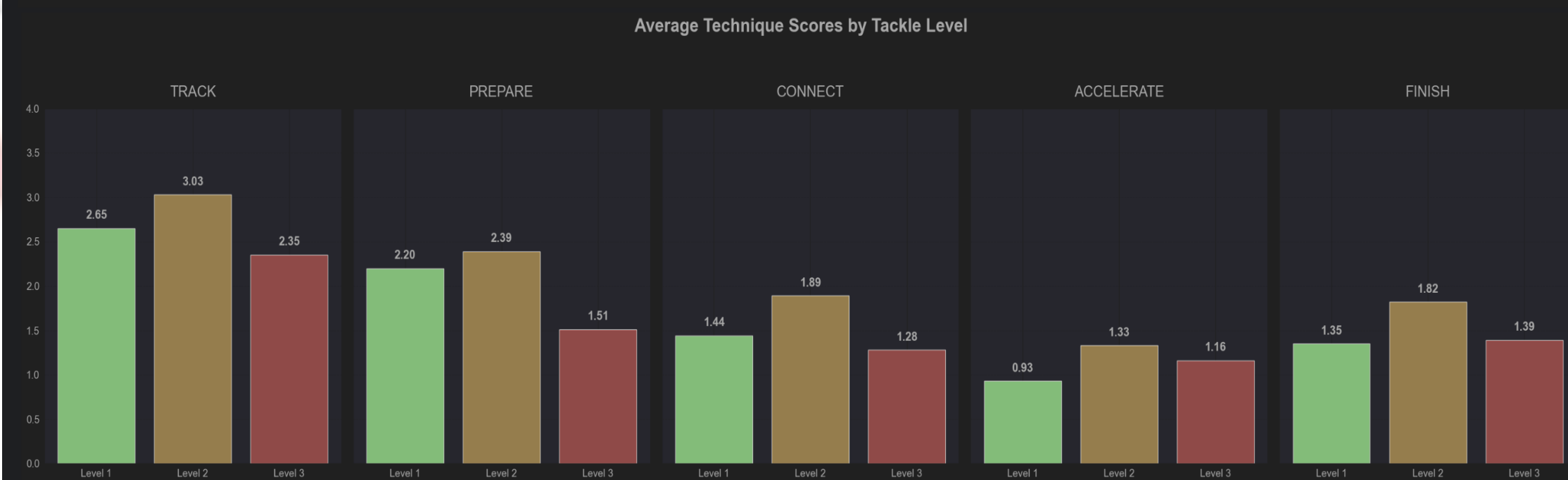


# **Visualizations: Bar Chart Technique AVGs**

- Weeks  
-1:6



- Weeks  
-1:10



# Comprehensive Analyses

## Chi-Square Test on Level 2 Vs. Level 3 and Explosive Plays

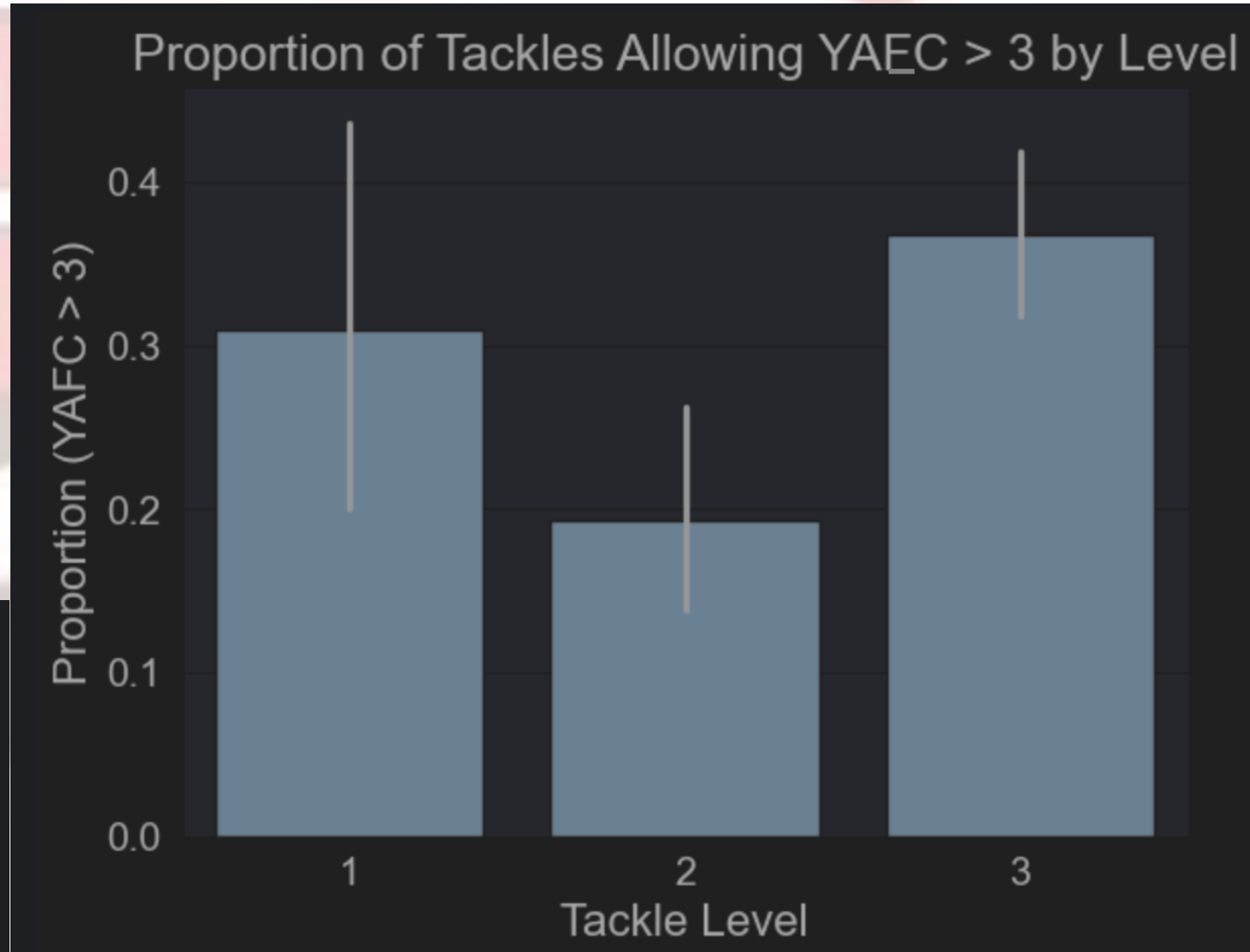
- Explosive Play =  $YAFC \geq 3$ 
  - $\{ 1: \geq 3$
  - $\{ 0: < 3$
- Checking the Independence between Tackle Level and  $YAFC \geq 3$

DF: 2, 14.5 is big, P-Val = 0.0007

Conclusion: Reject Null. There is a Statistically Significant

Relationship between Tackle Level and whether a tackle allows more than 3 YAFC.

| YAFC_gt3                        | 0   | 1   |
|---------------------------------|-----|-----|
| LEVEL                           |     |     |
| 1                               | 38  | 17  |
| 2                               | 117 | 28  |
| 3                               | 216 | 126 |
| Chi-square = 14.504, p = 0.0007 |     |     |



# Comprehensive Analyses

## Two-Sample Z Test on Level 2 vs Level 3 Make %

- Outcome is Binary, so Z Test
- $H_0$ : Make Probability is the same for Level 2 and Level 3 Tackles
- $H_a$ : Statistically significant difference in Make %
- Result: Reject  $H_0$ : Z-Score 2.99, P-Val < 0.05, Statistically significant results show that Level 2 tackles are made at higher rates than level 3.
- Make % by Level
  - { Level 2: 71.72%
  - { Level 3: 57.31%

|           | Level                          | Made Tackles | Total Tackles | Make %   |
|-----------|--------------------------------|--------------|---------------|----------|
| 0         | Level 2                        | 104          | 145           | 0.717241 |
| 1         | Level                          |              |               |          |
|           | 0 Level 2                      | 104          | 145           | 0.7172   |
|           | 1 Level 3                      | 196          | 342           | 0.5731   |
| Z-stat =  | Proportion Difference = 0.1441 |              |               |          |
|           | 95% CI = [0.0540, 0.2343]      |              |               |          |
|           | Z-stat = 2.9907                |              |               |          |
| P-value = | P-value = 0.00278353           |              |               |          |

# Comprehensive Analyses

## Two-Sample T-Test on MADE Level 2 vs Level 3 and AVG YAFC

- Only Made Tackles Considered
- $H_0$ : Among made tackles, Level 2 and 3 allow the same AVG YAFC
- $H_a$ : Statistically significant difference in each level's AVG YAFC Allowed
- AVG YAFC by Level
  - { Level 2: 1.221
  - { Level 3: 1.673
- Result: Reject  $H_0$ : T-stat = -2.664, P-Val < 0.05, Level 2 AVG YAFC is statistically significantly lower than Level 3 AVG YAFC

mean YAFC = 1.221

mean YAFC = 1.673

Mean difference = -0.452

t-test: t = -2.664, p-value = 0.00816969

NOTE: Used 'equal\_var=False' to not assume both groups have equal variance.

# Comprehensive Analyses

## Two-Sample T-Test on Level 3 vs Level 2 Technique Score

- $H_0$ : Mean TECH\_SCORE is the same between Level 3 and 2 Tackles
- $H_a$ : Statistically significant difference in each level's mean TECH\_SCORE
- Mean TECH\_SCORE by Level
  - { Level 3: 7.699
  - { Level 2: 10.469
- Result: Reject  $H_0$ : T-stat = -2.664, P-Val < 0.05, Level 3 AVG TECH\_SCORE is statistically significantly lower than Level 2 AVG TECH SCORE

```
Level 2 tackles: n = 145, mean TECH_SCORE = 10.469
Level 3 tackles: n = 342, mean TECH_SCORE = 7.699
Mean difference (L2 minus L3) = 2.770
t-test: t = -2.664, p-value = 0.0046089
```

NOTE: Used 'equal\_var=False' to not assume both groups have equal variance.

# Comprehensive Analyses

## Two-Sample Z Test on ONLY ENGAGED Level 2 vs Level 3 Tackles Make %

- Outcome is Binary, so Z Test
- $H_0$ : Make Probability is the same for ENGAGED Level 2 and Level 3 Tackles
- $H_a$ : Statistically significant difference in ENGAGED MAKE %
- ENGAGED MAKE % by Level
  - { Level 2: 53.57%
  - { Level 3: 50.43%
- Result: Reject  $H_0$ : Z-Score 0.2977, P-Val > 0.05, Cannot reject  $H_0$
- Football Explanation

### Two-proportion z-test: MAKE% (Engaged only)

|   | Group           | Made Tackles | Total Tackles | Make % |
|---|-----------------|--------------|---------------|--------|
| 0 | Engaged Level 2 | 15           | 28            | 0.5357 |
| 1 | Engaged Level 3 | 58           | 115           | 0.5043 |

Z-stat = 0.2977

P-value = 0.765894

# Comprehensive Analyses

## Two-Sample Z Test on ONLY NON- ENGAGED Level 2 vs Level 3 Tackles Make %

- Outcome is Binary, so Z Test
- $H_0$ : Make Probability is the same for NON - ENGAGED Level 2 and Level 3 Tackles
- $H_a$ : Statistically significant difference in NON-ENGAGED MAKE %
- NON-ENGAGED MAKE % by Level
  - { Level 2: 76.07%
  - { Level 3: 60.79%
- Result: Reject  $H_0$ : Z-Score 2.8332, P-Val < 0.05, reject  $H_0$ . There is a Statistically Significant difference in Make % between Non-Engaged Level 2 and Level 3 Tackles
- Football Explanation

Two-proportion z-test: MAKE% (Not engaged only)

|   | Group               | Made Tackles | Total Tackles | Make % |
|---|---------------------|--------------|---------------|--------|
| 0 | Not Engaged Level 2 | 89           | 117           | 0.7607 |
| 1 | Not Engaged Level 3 | 138          | 227           | 0.6079 |

Z-stat = 2.8332

P-value = 0.0046089

# Conclusion...

- Level 2 Tackles are more efficient than Level 3 Tackles!
- Next steps: Predicting Tackles
- How?
  - Binomial Distribution (Beta-Binomial)
  - Machine Learning
  - Test on NFL DATA (will require tracking data to make this happen)